LAB1

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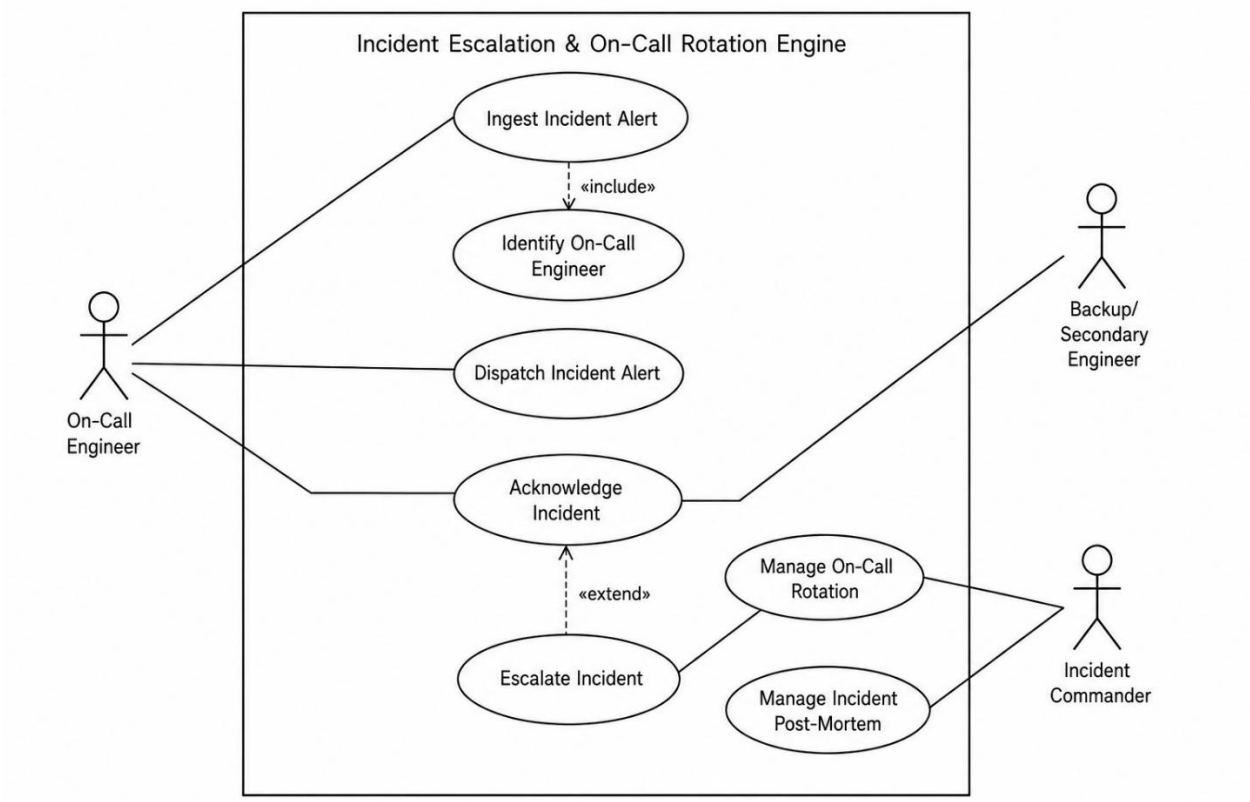
Requirements Table

	Type	Description	Priority	Acceptance Criteria	Rationale (short)	Comments
FR-0 01	Functional	The system shall ingest incident alerts, identify the active on-call engineer from the rotation schedule, and escalate the alert to the secondary engineer if it is not acknowledged within 5 minutes.	High	**Pass:** Escalation is triggered after 5 minutes when the incident is not acknowledged. **Fail:** An unacknowledged P1 incident is not escalated.	Ensures critical incidents are not missed when the primary on-call engineer does not respond.	
FR-0 02	Functional	The system shall dispatch/send incident alerts to the assigned on-call engineer through the configured notification channels such as phone, SMS, or email.	High	**Pass:** The assigned engineer receives the alert through the configured channel after an incident is ingested. **Fail:** The alert is not dispatched through the configured channel.	Ensures that on-call engineers are immediately informed about incidents.	
FR-0 03	Functional	The system shall maintain the on-call rotation schedule and identify the	High	**Pass:** For any given shift, the system displays	Ensures incidents are mapped to the correct engineer	

		engineer assigned to the current shift		the correctly assigned on-call engineer. **Fail:** The system displays an incorrect or unavailable engineer.	during each shift.	
FR-0 04	Functional	The system shall allow the notified on-call engineer to acknowledge an assigned incident	High	**Pass:** The incident status changes to “Acknowledged” when the engineer confirms the alert. **Fail:** The incident remains unacknowledged after confirmation.	Allows the system to determine whether escalation is required.	
FR-0 05	Functional	The system should allow the Incident Commander to create and update a post-mortem record for a resolved incident.	Medium	Medium**Pass:** A post-mortem record can be created and updated for a resolved incident and is linked to its incident record. **Fail:** a system does not save or link the post-mortem recordcorrectly.	Helps the team document and review incidents after they are resolved.	
NFR001	Nonfunctional(Performance & Security)	The system shall initiate alert dispatch through webhook, SMS, and email within 3 seconds of	High	**Pass:** Benchmarking confirms alert dispatch starts within 3 seconds under simulated peak load while meeting the required security	Ensures alerts reach the responsible engineer quickly during incident	

		alert ingestion.		standards. **Fail:** Dispatch takes more than 3 seconds or security requirements are not met.		
NFR002	Nonfunctional	The system shall remain available for incident alert processing and escalation at all time.	High	**Pass:** Monthly monitoring shows system availability of at least 99.9%. **Fail:** Availability falls below 99.9%.	Prevents missed or delayed escalations caused by system downtime.	

UML Use-Case Diagram



Actors:

- On-Call Engineer: Receives incident alerts and acknowledges incidents.
- Backup/Secondary Engineer: Receives escalated incidents and acknowledges them.
- Incident Commander: Manages on-call rotations and incident post-mortems.

Use Cases:

- Ingest Incident Alert: Receives an incident alert into the system.
- Identify On-Call Engineer: Identifies the currently active engineer from the rotation schedule.
- Dispatch Incident Alert: Sends the incident alert to the responsible engineer.
- Acknowledge Incident: Allows the engineer to acknowledge the received incident.
- Escalate Incident: Escalates the incident to the Backup/Secondary Engineer when it is not acknowledged within 5 minutes.
- Manage On-Call Rotation: Maintains and updates the team's on-call shift schedule.
- Manage Incident Post-Mortem: Allows the Incident Commander to record and manage information about a completed incident.

Use-Case Flow

Goal	Escalate Incident
Actor	On-Call Engineer, Backup/Secondary Engineer
Preconditions	1. An incident alert has been received by the system.2. The active On-Call Engineer has been identified.3. The incident has been dispatched to the On-Call Engineer.
Postconditions	1. The incident is escalated to the Backup/Secondary Engineer if not acknowledged within 5 minutes.2. The escalation status is recorded in the system.
Main Success Scenario	**1.** The system receives an incident alert.**2.** The system identifies the active On-Call Engineer.**3.** The system dispatches the alert to the On-Call Engineer.**4.** The system waits for acknowledgement.**5.** The 5-minute period expires without acknowledgement.**6.** The system triggers escalation.**7.** The system identifies the Backup/Secondary Engineer.**8.** The system dispatches the alert to the Backup/Secondary Engineer.**9.** The Backup/Secondary Engineer acknowledges the incident.**10.** The system records the incident as escalated and acknowledged.
Alternate Flow	**On-Call Engineer Acknowledges Incident:** At Step 5, if the On-Call Engineer acknowledges within 5 minutes, the system records the incident as **Acknowledged** and does not trigger escalation.